

physics modelling agent

progress report

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supervisor 2: Prof. Dr. Samir Lounis

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outline

- 1 goal of the project
- 2 key definitions
- 3 architecture
 - constraints
 - crewai
- 4 design process
 - challenges during the design process
- 5 example workflow: The Edelstein Effect
- 6 next steps
- 7 summary

aim of the project

what the “physics modelling agent” should be

A configuration of “intelligent systems” which assists a human researcher in building a model to answer a research question.

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- “physics modelling agent” takes the input:
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reports findings:

- human readable **model description**
- executable **numerical implementation**

definitions of LLM and agentic AI

what is an LLM

- “Transformer language models” [19] with at least billions of parameters (no clear cut-off) [19]
- mostly use tokenization: dividing text into pieces and applying transformation function

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what is an agentic AI

- an AI with enhanced capabilities including: strategic planning, (external) tools usage, collaboration with other agents [3]

architectural foundations

constraints

- only use open source software
- try to use only free software (no payed subscriptions)

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- Scalable AI Accelerator (**SAIA**) from the GWDG[10]
- access via remote servers with valid API key [9]

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design choices

- use agentic AI to harness strategic planning capabilities and tool usage
- Scalable AI Accelerator (**SAIA**) from the GWDG[10]
- access via remote servers with valid API key [9]
 - use an existing framework for AI orchestration
 - save time on the software engineering part of the project
 - focus on definition of AI agents and configuration of their behaviour



Figure: framework chosen: crewai [6]

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accesses an agentic AI model to perform a task

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- allows to access any AI-models we have access to (SAIA)
- behaviour can be defined via parameters (role, goal, backstory, creativity)
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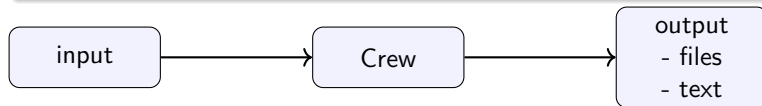
- defined by task description and expected output
- is assigned an agent (execution merges agent and task description)

the Crew[5]

contains all agents with their tasks and handles the interactions between agents

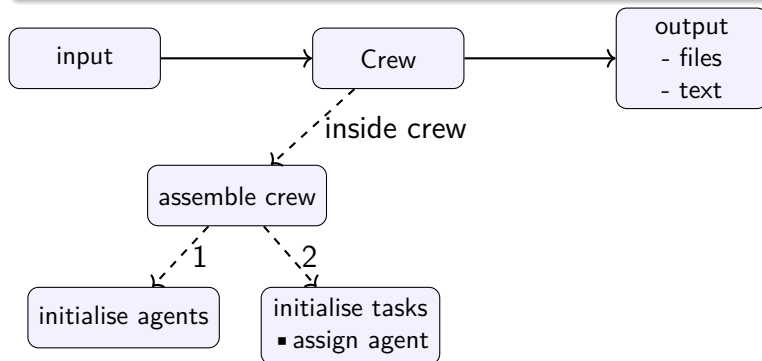
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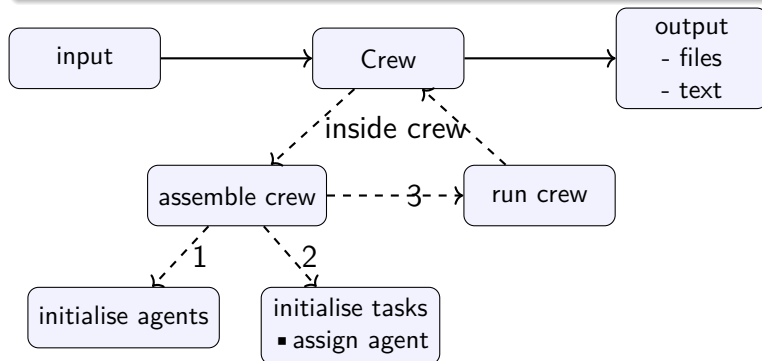
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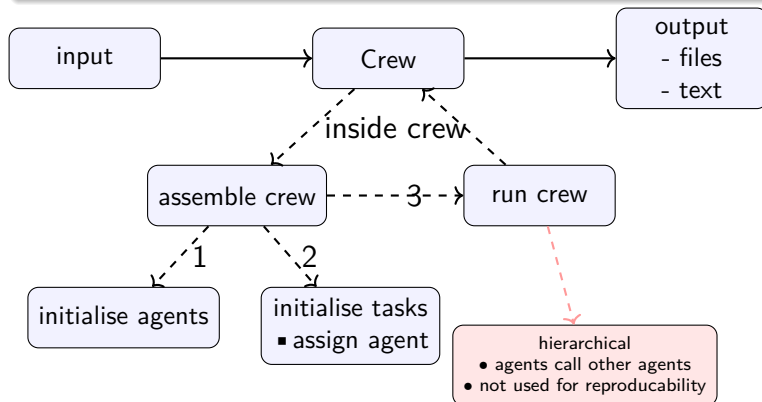
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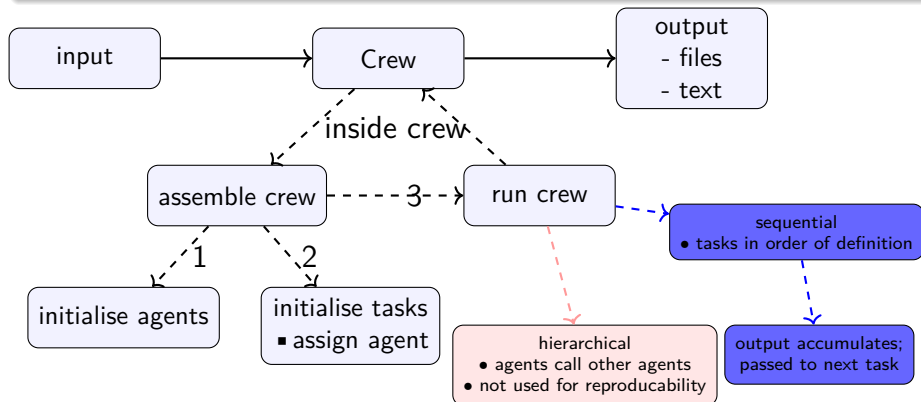
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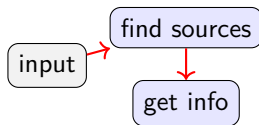
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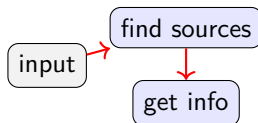
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information finding

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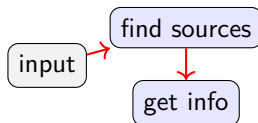
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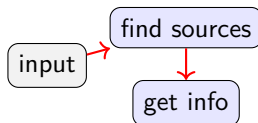
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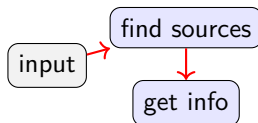
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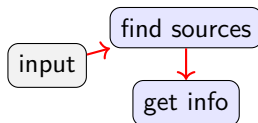
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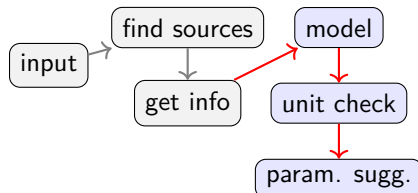
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 - ✓ implement custom tools performing back-off strategy
 - ✓ separate the task performing the arXiv request from the crew
 - ⇒ no new requests when testing other tasks



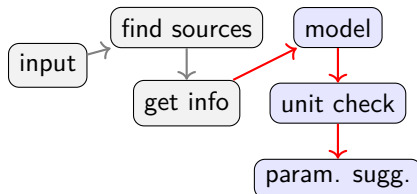
the unit problem

- 1 task outputs formula from the sources; theoretical formulas often use different definitions of parameters



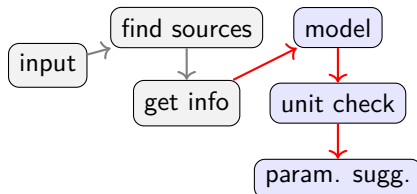
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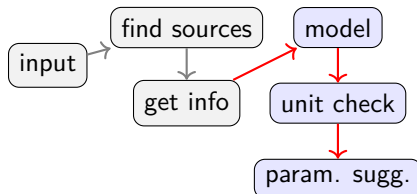
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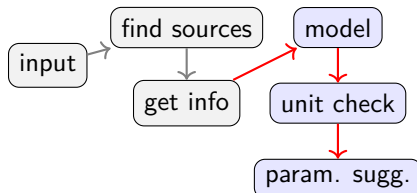
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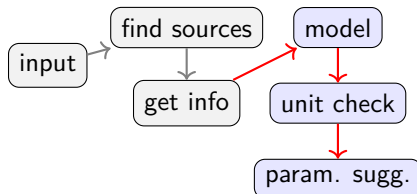
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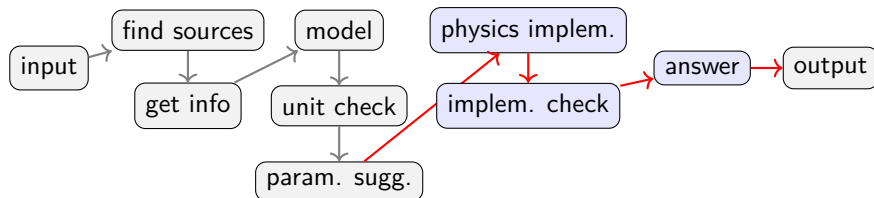
- for realistic results realistic start parameters are necessary
- ⇒ agent for suggesting reasonable starting parameters



numerical implementation

minor challenges

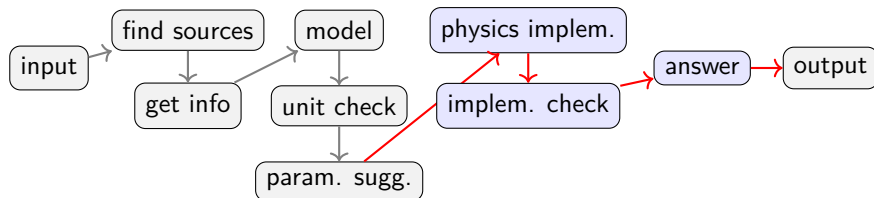
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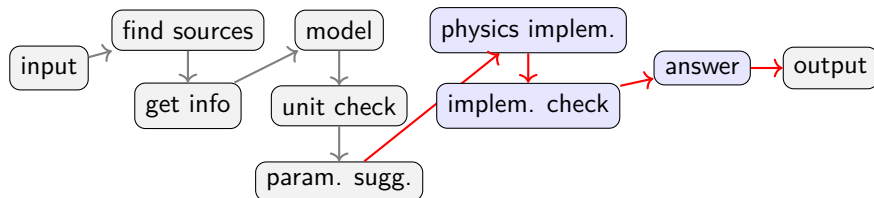
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answer: provide a short result at runtime

- needed for automatic evaluation of the system (benchmarking)



example: input/modelling request

input

aim : “Calculate the Edelstein effect for a Rashba fermion (at the Gamma point of the Brillouin zone).

Compute the magnetization magnitude and direction of different directions and magnitudes of the applied electric field.

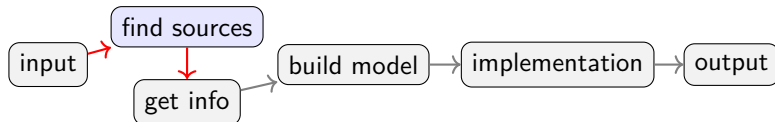
Consider how the result depends on relevant parameters of the model (e.g. chirality, fermi velocity, spin-orbit coupling strength) and make explicit graphics.”

topic : Edelstein-Effect

papers found

agent configuration:

- no creativity (temperature = 0)
- LLM model: qwen3.5-122b-a10b[16] (general purpose & usage)
- tools: arXiv_Papers_With_Backoff



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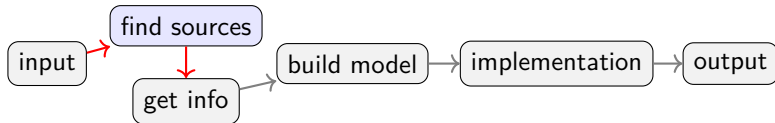
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papers found ([view full report](#))

“Edelstein Effect in Isotropic and Anisotropic Rashba Models”[12]

“Spin and orbital Edelstein effect in spin-orbit coupled
noncentrosymmetric superconductors”[1]

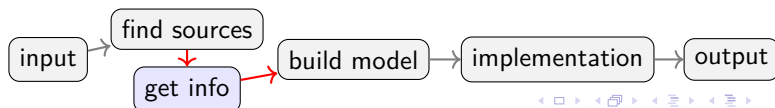
“Spin and orbital Edelstein effect in a bilayer system with Rashba
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extracted information ([view full report](#))

$$\hat{H} = \frac{p^2}{2m} + \alpha \hat{z} \cdot (\vec{p} \times \vec{\sigma})$$

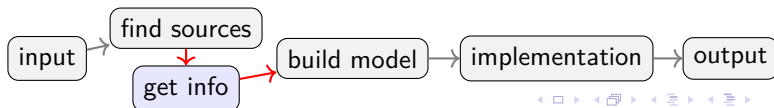
Rashba-Hamiltonian

$$M_y = \frac{\mu_b |e| \tau}{2\pi} m \alpha [\hat{z} \times \mathbf{E}]_y$$

magnetization in HDR

$$M_y = \frac{\mu_b |e| \tau}{2\pi} \sqrt{m^2 \alpha^2 + 2m E_F} [\hat{z} \times \mathbf{E}]_y$$

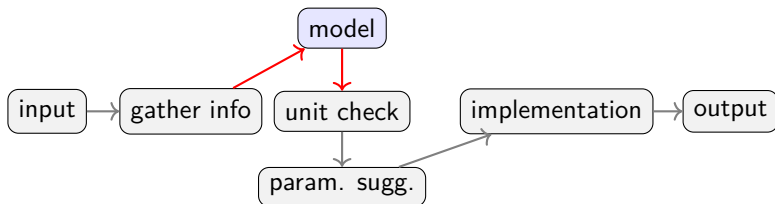
magnetization in LDR



build model

agent configuration:

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- LLM model: deepseek-r1-distill-llama-70b[13]; (optimized for logic & reasoning)
- tools: this LLM is unable to use tools



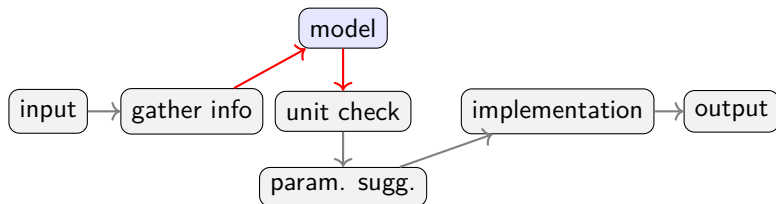
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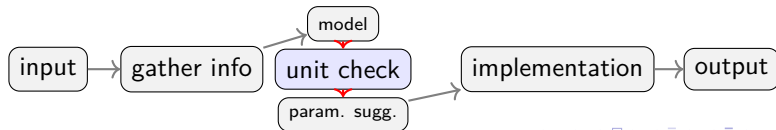
extracted information ([view full report](#))

restates results from information extraction



unit check

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unit check

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unit corrections ([view full report](#))

$$\hat{H} = \frac{p^2}{2m} + \frac{\alpha}{\hbar} \hat{z} \cdot (\vec{p} \times \vec{\sigma})$$

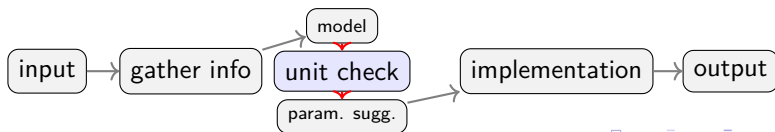
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magnetization in HDR

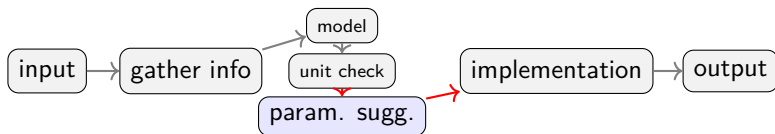
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magnetization in LDR



parameter suggestion

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suggested starting parameters ([view full report](#))

$$m \approx 6,10 \times 10^{-32} \text{kg}$$

effective mass

$$\alpha = 5,0 \times 10^4 \text{m/s}$$

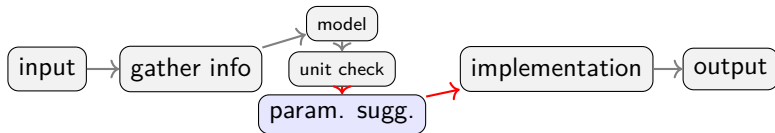
Rashba Coupling Strength

$$\tau = 5,0 \times 10^{-13} \text{s}$$

transport time

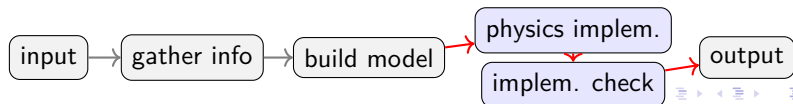
$$E_x = 1,0 \times 10^5 \text{V/m}$$

applied electric field



numerical implementation I

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devstral-2-123b-instruct-2512 [11]
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numerical implementation I

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results for magnetisation ([view full report](#))

Isotropic HDR Magnetization:

$$M_y = 3,25 \times 10^{-9} \text{ A/m}$$

Isotropic LDR Magnetization:

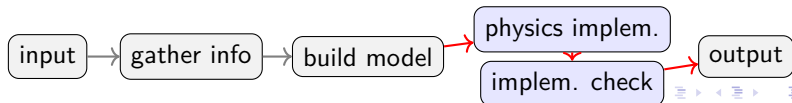
$$M_y = 1,52 \times 10^{-8} \text{ A/m}$$

Anisotropic Magnetization Components:

$$M_x = 2,04 \times 10^{-8} \text{ A/m}$$

$$M_y = 2,04 \times 10^{-8} \text{ A/m}$$

some minor corrections necessary



numerical implementation II

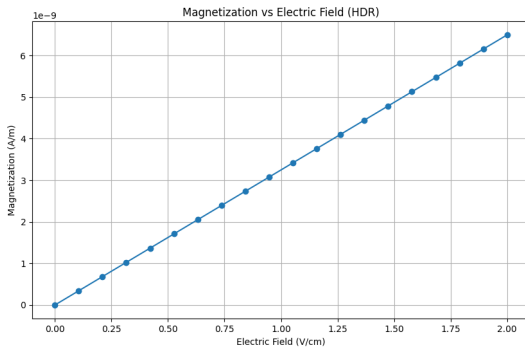


Figure: magnetisation over electric field



numerical implementation III

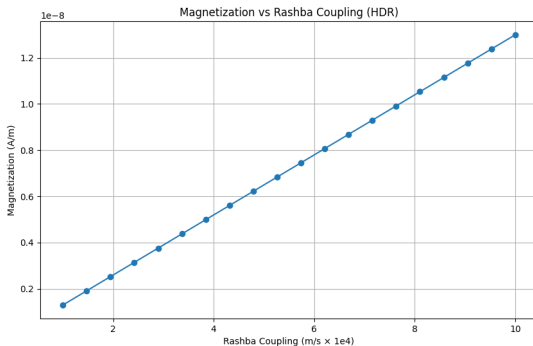


Figure: magnetisation over Rashba param. α



next steps

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a method to measure the systems performance

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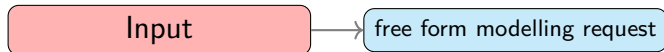
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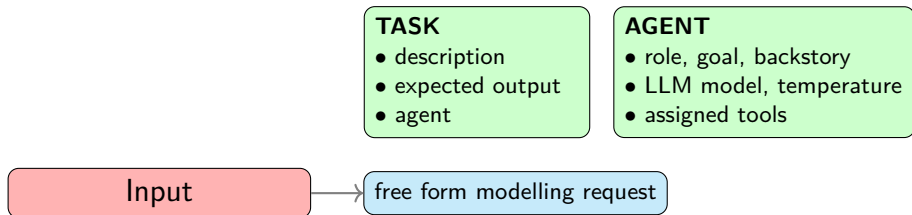
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- ⑤ sources: how does the number (and possibly type) of sources influence the systems results?

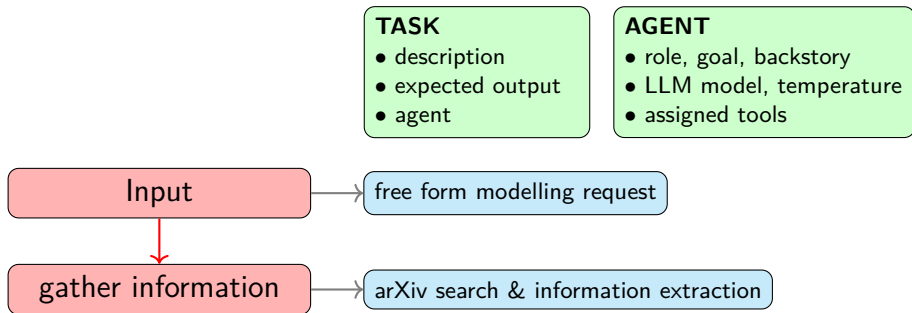
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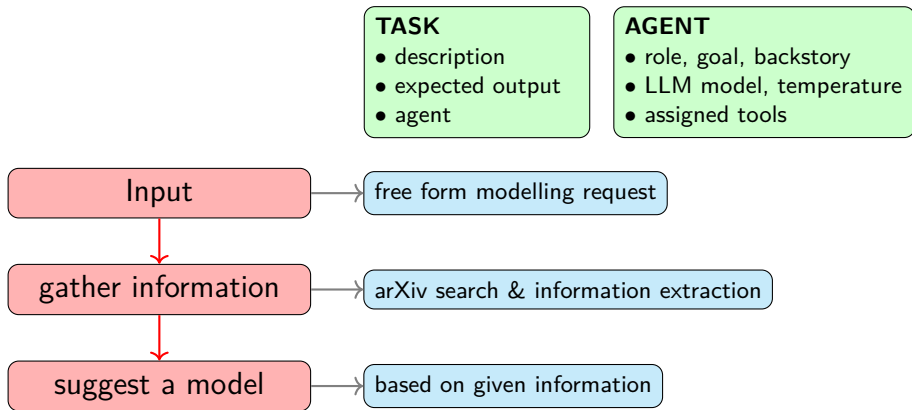
summary



summary



summary



summary

TASK

- description
- expected output
- agent

AGENT

- role, goal, backstory
- LLM model, temperature
- assigned tools

Input

free form modelling request

gather information

arXiv search & information extraction

suggest a model

based on given information

check & correct units

dimensional analysis

summary

TASK

- description
- expected output
- agent

AGENT

- role, goal, backstory
- LLM model, temperature
- assigned tools

Input

free form modelling request

gather information

arXiv search & information extraction

suggest a model

based on given information

check & correct units

dimensional analysis

implement model

executable python code calculations

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